

Peter N. Dobbs

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PROFESSIONAL EXPERIENCE GLOBAL ORDNANCE, SARASOTA, FL

AI Engineer (Lead)

December 2025 - present

- Architecting cloud-based data enrichment pipeline for sporting goods e-commerce
- Implementing software development lifecycle practices to comply with ISO 27001
- Mentoring junior software engineer

CANON MEDICAL INFORMATICS, INC, MINNETONKA, MN

Senior Product Analyst

February 2025 - May 2025

- Highly visible role with end-to-end ownership of platform development across global teams (Minnesota, Japan, Edinburgh)
- Leading live customer experience feedback collection and experimentation
- Making data-driven decisions based on product usage and performance metrics
- Collaborating with SMEs/KOLs to share technical and medical industry knowledge with product teams

Software Engineer

February 2022 - February 2025

- Championed projects to enhance integration with external data sources
- Implemented compliance to the latest Cybersecurity Guidance from the FDA
- Drove collection and analysis of product usage and performance data
- Collaborated with Product Managers and various Stakeholders across the globe to construct product roadmaps
- Led defect review board and product hazard analysis activities
- Defined and implemented new processes for handling vulnerabilities reported against an SBOM

Associate Software Engineer

May 2020 - February 2022

- Designed and implemented data model for financial and operational analysis of medical imaging departments
- Developed custom Apache NiFi processors to drive ETL of HL7 and DICOM medical data
- Created visualization of hospital operations with Elasticsearch and Kibana
- Implemented ATNA-based auditing for Role-based Access Controls
- Designed data removal request process for GDPR-compliance

MARQUETTE ENERGY ANALYTICS, LLC, MILWAUKEE, WI

Software & Data Generalist — Product Owner

October 2018 - May 2020

- Primary contact for licensed products at five of our top customer sites
- Designed custom web data layer and C# Excel add-in for performing advanced analysis of energy demand forecasts
- Deploying regular updates for forecasting models on Azure to cover 30% of nation's energy demand
- Culture-influencer within the startup

EDUCATION

Master of Science, Mathematical, Statistical, and Computational Sciences

Marquette University Graduate School, Milwaukee, WI

Bachelor of Science, Biomedical Engineering - Biocomputing

Marquette University Opus College of Engineering, Milwaukee, WI

APPLIED SKILLS

Programming Languages: Python, SQL, R, Matlab, C#, Java
Data Analytics: PyTorch/TensorFlow, Jupyter Notebooks, RStudio, PowerBI, Apache NiFi, Elasticsearch/Kibana, Grafana, Apache Spark, LangChain
Development: Git, Agile Scrum, Docker, CI/CD, Deon Ethical Checklist, REST API, Data Layers, Data Orchestration, Data Warehousing, Data Modeling
CySec: Risk Analysis, CVSS, EPSS, VEX, SBOM, CycloneDX, DependencyTrack, ATNA auditing
Platforms: Mac OS X, Windows, Linux, AWS, Azure
Defense Contractor Experience: ITAR
SaMD Experience: 21 CFR 820, ISO 13485:2016, EU MDR 2017/745, ISO 62366-1
Health Standards Experience: HL7v2, HL7 FHIR, DICOM, DICOMweb, IHE

FEATURED PROJECTS

[peterdobbs77](#)

Agentic Healthcare Analytics [peterdobbs77/agentic-healthcare-analytics](#)
WIP Agentic AI system for performing analytics on healthcare data. Providing an agentic interface for interacting with FHIR Resources.

Daily Coding [peterdobbs77/daily-coding](#)
Keeping my skills fresh and learning new ones. Lately I've been upskilling in Generative AI and Machine Learning.

Object Tracking [peterdobbs77/objectTracking.py](#)
Experiments with Computer Vision to perform object detection and tracking. The main focus is on analyzing film of ultimate frisbee games, but I also have been extending my experience with eye tracking (continuing from my Eye Tracking project).

EM Algorithm [peterdobbs77/shiny-em-algorithm](#)
The Expectation Maximization Algorithm is an iterative method that can be used to find model parameters. This project implemented a R shiny app that takes in data from a 1-dimensional mixture model and finds the optimal parameters to represent the distribution of that data. The final product is published to [peterdobbs.shinyapps.io/em-algorithm](#).

Team Great Lakes (.tgl) - Team Lead [GreatLakesSIIM](#)
Led a team of students from Milwaukee and Chicago in the Society of Imaging Informatics in Medicine (SIIM) Hackathon at the 2018 and 2019 SIIM Annual Meeting. As Team Lead, I was directly involved in the requirements gathering and development of projects that won the hackathon in 2018 and placed third in 2019.

- **Forms on FHIR (2019)** - provides a UI for directly creating and editing FHIR resources for populating an AI-ready research FHIR server.
- **Follow-up of Noncritical Actionable Findings (2019)** - automated notification of primary care providers that important findings which are not critical for patient intervention at the time of the study are contained in radiology study. *This project placed 3rd at the 2019 SIIM Hackathon.*
- **Synoptic Reporting as an Enabler (2018)** - converts a synoptic (structured and coded) radiology report into a FHIR resource for storage in a medical research database for AI use cases. *This project was part of a series of presentations that placed 1st at the 2018 SIIM Hackathon.*

CERTIFICATES

Agentic AI [Course Certificate](#) DeepLearning.AI 9 October 2025
Developing multi-agent systems for project planning, data analysis and visualization, customer support, market research, and more

CVSS v3.1: FIRST Learning 13 December 2024
Detecting and scoring security vulnerabilities according to the standard CVSS process.

A-CSPO: Advanced Certified Scrum Product Owner
CSPO: Certified Scrum Product Owner

21 November 2023
6 August 2021

QI 104: Interpreting (Healthcare Quality) Data

29 October 2018

PUBLICATIONS **Towards Developing an EMR in Mental Health Care for Children's Mental Health Development among the Underserved Communities in USA.** 2021
Kazi Zawad Arefin, Kazi Shafiul Alam, Masud Rabbani, **Peter Dobbs**, Leah Jepson, Amy Leventhal, Amy Van Hecke and Sheikh Iqbal Ahamed
doi: 10.1109/COMPSAC51774.2021.00100

FEATURED TALKS

SIIM Clinical Data Informatics

DICOM For Informaticists, Part 1

DICOM For Informaticists, Part 2

Pair of recorded lectures, created as a part of the SIIM CDI Series, introducing the DICOM standard to aspiring Clinical Data Informaticists.



SIIMcast Episode 49: Hackathon 2

In Spring of 2020, I was invited to speak about my experiences at the SIIM Hackathon and advocate for others to participate.

Nyandwi Muzungu: Medical Device Repair in Constrained Environments

In November 2019, I was invited to give a talk as a part of the Global Innovation Seminar Series at University of Wisconsin. The event coordinators asked me to share some of my experiences from my trips to East Africa.



REFERENCES available upon request